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Studierichting: Informatica
Oefeningenreeks 2E nr. 5.5

$$\lim_{x\to 0}\left(\frac{ax+b}{cx+b}\right)^{\frac{1}{x}}$$

$$\left(s=\frac{1}{x}\right)$$

$$=\lim_{s\to\infty}\left(\frac{as^{-1}+b}{cx^{-1}+b}\right)^{s}$$

$$=\lim_{s\to\infty}\left(\frac{bs+a}{bs+c}\right)^{s}$$

$$(z=bs+c)$$

$$=\lim_{z\to\infty}\left(\frac{z-c+a}{z}\right)^{\frac{z-c}{b}}$$

$$=\lim_{z\to\infty}\left(1+\frac{a-c}{z}\right)^{\frac{z}{b}}\cdot\lim_{z\to\infty}\left(1+\frac{a-c}{z}\right)^{\frac{-c}{b}}$$

$$=\left(\lim_{z\to\infty}\left(\frac{z-c+a}{z}\right)^{\frac{z}{a-c}}\right)^{\frac{a-c}{b}}\cdot 1$$

$$=e^{\frac{a-c}{b}}$$

References